

Strengthening STEM Education Through Virtual Reality



Universidad
Nacional
de San Martín



Immersive STEM Labs



Teacher Training



Measurable Pilots



A science + education + immersive development partnership.

UNSAM (National University of San Martín)



A public university with strong scientific research capacity and a commitment to educational innovation.
UNSAM provides curriculum alignment, scientific content expertise, and evaluation design.



Lupic Studio (Development Partner)



An immersive content and interactive development studio supporting the production of XR learning experiences, technical implementation, and deployment planning for school environments.



Together

We connect scientific knowledge, pedagogy, and immersive technology to deliver measurable improvements in STEM learning outcomes.

Logline

We are building immersive VR science laboratories for public secondary schools in Buenos Aires Province, enabling students to safely experiment with complex STEM concepts that are often inaccessible in traditional classrooms. The program combines Unreal-powered VR experiences, teacher training, and a measurable pilot in two schools to validate learning outcomes and scale regionally.



Hands-on STEM learning is not equally accessible.

Many public secondary schools lack laboratory infrastructure, time, and resources for high-quality experimentation. As a result, STEM concepts are often taught abstractly—reducing engagement, comprehension, and confidence. This gap is most severe in low-resource contexts, widening educational inequalities.



VR makes the invisible visible and practice possible.

-  Turns abstract phenomena (waves, atoms, viruses) into interactive experiences
-  Safe, repeatable experiments without physical lab constraints
-  Standalone headsets enable scalable deployment
-  High engagement + stronger concept retention through embodied learning



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A complete immersive **STEM** learning model. Not just an app

We deliver an end-to-end approach: Unreal-built VR labs, a teacher enablement program, classroom integration materials, and two pilot implementations with measurable evaluation.



VR Labs



Teacher



Pilot &



Scale

Program pillars:

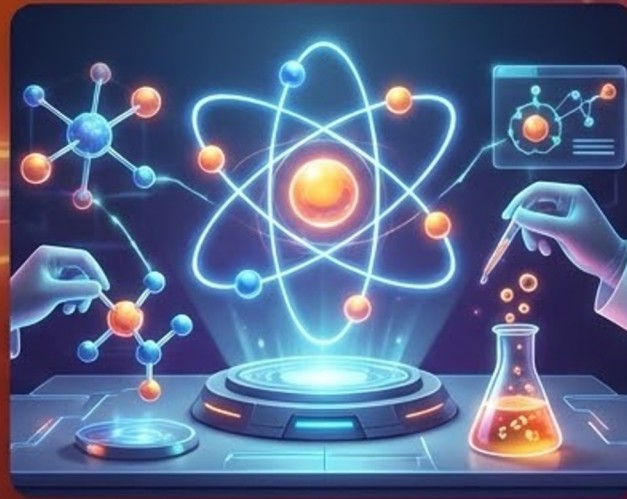
- 1 VR Laboratories (Physics, Chemistry, Biology) 
- 2 Teacher & school leadership training 
- 3 Manuals + classroom integration toolkit 
- 4 Pilot deployment + evidence generation 

Three immersive labs aligned with secondary STEM curricula



Physics — Wave Nature of Light

Explore interference, diffraction, wavelength, frequency, and propagation through interactive experiments.



Chemistry — Atomic Manipulation

Build atoms and molecules, visualize structure, and simulate reactions to understand matter and change.



Biology — Virus Hunters

Learn how viruses spread and how vaccines work, based on real research and public health relevance.

Designed for real classrooms and real teachers

Our instructional model supports teachers with structured activities before and after VR sessions, turning immersive experiences into measurable learning.



Frame the concept, introduce guiding questions, and activate prior knowledge.



Students actively explore, manipulate virtual objects, and conduct safe experiments.



Reflect on the experience, consolidate learning through discussion, and complete short assessments.



Teacher guides and lab manuals support adoption at scale.

Simple interactions. Strong pedagogy.

- Guided tasks with minimal interface complexity
- Immediate feedback through visual cause-effect
- Comfort-first VR design (locomotion options)
- Teacher-led flow, student-centered discovery



Who we serve in the pilot phase.

 Public secondary schools in San Martín district, Buenos Aires

 20 teachers & school leaders trained

 120 students reached through immersive STEM sessions

 Two pilot schools to validate results in real conditions



Clear impact targets with measurable evidence.

Impact objectives:



Increased conceptual understanding in targeted STEM topics



Higher student engagement and confidence in science learning



Teacher capacity to integrate immersive learning sustainably



A replicable toolkit for broader adoption in public education

KPIs (pilot):



Learning gain: pre/post concept checks per lab



Engagement: student surveys (interest, confidence, relevance)



Teacher adoption: training completion + integration rubric



Usability: VR comfort + session reliability metrics



Evidence-driven evaluation built into the pilot

Methods:



Pre/Post Learning Checks: short concept quizzes aligned to each lab



Teacher Observation Rubrics: engagement, collaboration, misconceptions



Student Feedback: clarity, confidence, motivation toward STEM



Implementation Metrics: session duration, stability, usability issues



Pilot Report: findings, lessons learned, iteration roadmap

A background image of a classroom. In the foreground, a young boy with dark skin and curly hair is sitting at a desk, looking at a tablet. A teacher, a woman with dark curly hair, is leaning over him, looking at the tablet. In the background, other students are visible, some looking at tablets. The room has large windows on the left side.

Equity-first deployment in public education.

This initiative is designed to reduce opportunity gaps by bringing high-quality, hands-on STEM learning to public schools where lab infrastructure is limited. We prioritize accessibility, teacher support, and low-barrier deployment to ensure students who benefit most can participate fully.

Commitments:

Public-school focus and pilot access at no cost to students

Accessible design choices (comfort, usability, inclusive participation)

Teacher empowerment to sustain the model beyond the grant



Public-school focus on design choices (comfort, usability, inclusive)



Accessible deployment participation



Teacher empowerment to sustain the model beyond the grant



Aligned with the UN Sustainable Development Goals



SDG 4 — Quality Education



SDG 9 — Industry, Innovation & Infrastructure



SDG 10 — Reduced Inequalities



SDG 17 — Partnerships

Why Unity: scale-ready delivery for schools.

Our current prototypes were developed using Unreal Engine. To scale reliably across standalone VR devices and school environments, we will establish a dedicated Unity development team and production pipeline led by Lupic Studio, in collaboration with UNSAM's scientific and educational stakeholders.

What Unity unlocks:

-  Optimized performance for standalone VR hardware
-  Modular pipeline for multiple labs and future expansions
-  Faster iteration informed by pilot evaluation results
-  Sustainable maintenance and broader distribution options



Technical approach (Unity)



Unity-based VR development using modern XR tooling



Standalone VR headset deployment for school settings



Modular scenes per discipline for updates and extensions



Data collection hooks for evaluation metrics



Accessibility and comfort settings built in from day one



Challenges we will address



Performance optimization
on standalone devices



First-time VR user
onboarding in classrooms



Stability and
repeatability for
teachers



6-8 month implementation plan.



Curriculum
alignment +
scientific scripting
(UNSAM lead)



Unity pipeline
setup + lab
development
(Lupic lead)



Internal testing +
usability
refinement
(joint)



Teacher training +
manuals
finalization
(UNSAM lead)



Pilot deployment
(2 schools)
(joint)



Evaluation +
dissemination
scale plan
(UNSAM lead)



Month 1



Month 2



Month 3



Month 4



Month 5



Month 6

Budget request: \$100,000 USD

Allocation (categories):



This grant funds the Unity-based production and pilot validation needed to prove impact and build a scalable model for public education.

A partnership built for impact and scalability.

UNSAM contributes scientific expertise, curriculum alignment, teacher training, and evaluation design. **Lupic Studio** leads immersive production and the Unity development pipeline required for scale-ready deployment. Together, we will deliver **measurable STEM learning outcomes** through VR in real public-school contexts.

With Unity for Humanity support, we will expand access to high-quality science learning—helping students engage with **STEM** through immersive, equitable, and evidence-driven education.

